

The empirical recurrence for OEIS A254737, recovered from its tail

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Abstract

OEIS A254737 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 181$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 63$. Its last coefficient is $c_{57} = -402653184 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254737 is “Number of $(n+2) \times (2+2)$ 0..1 arrays with every 3×3 subblock sum of the two medians of the central row and column minus the sum of the minimums of the diagonal and antidiagonal nondecreasing horizontally and vertically.”. It has offset 1 and begins

2848, 19782, 130554, 795666, 4613180, 26334567, 148992766, 835291656, ...

The entry states:

Empirical recurrence of order 57 (see link above).

As of the “Last modified” line on the live entry (Jun 19 2025 20:59:53, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 181$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 181$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 362$ exact terms from $n = 6$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	11	c_{16}	−1237017	c_{31}	19092181377	c_{46}	−39467554816
c_2	−27	c_{17}	−6783575	c_{32}	−345081400	c_{47}	16403734528
c_3	−55	c_{18}	−14392510	c_{33}	−5915548456	c_{48}	15790104576
c_4	302	c_{19}	59693809	c_{34}	7640092113	c_{49}	−6874693632
c_5	−1287	c_{20}	44744466	c_{35}	−41358661182	c_{50}	27295744
c_6	5444	c_{21}	−19571	c_{36}	29094958450	c_{51}	−4819517440
c_7	−7793	c_{22}	6923240	c_{37}	37530762060	c_{52}	3879469056
c_8	5938	c_{23}	−502115483	c_{38}	−49796359528	c_{53}	177209344
c_9	−3395	c_{24}	−437856448	c_{39}	36694638640	c_{54}	−2963275776
c_{10}	−123298	c_{25}	886252415	c_{40}	−33665725680	c_{55}	1652555776
c_{11}	233263	c_{26}	1224723402	c_{41}	−38504136928	c_{56}	503316480
c_{12}	−148241	c_{27}	661792021	c_{42}	89555727520	c_{57}	−402653184
c_{13}	577370	c_{28}	875162287	c_{43}	−27880521408		
c_{14}	1588020	c_{29}	−7731802900	c_{44}	−15737712768		
c_{15}	−4006053	c_{30}	−6214649528	c_{45}	20840841728		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 63$, and it is the recurrence OEIS A254737 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{56} - \dots - c_{57}$. Its constant term is $-c_{57} = 402653184$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -402653184 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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